Problem Set 2: Volatility

MGT 927

Data and methods. CRSP daily data from WRDS, 2 Jan 1990 – 31 Dec 2020 (7,812 trading days). The market is the CRSP value-weighted return (`vwretd`); MSFT is its CRSP return. Returns are in percent per day. All likelihoods are coded by hand and maximized numerically. Standard errors are robust (Bollerslev–Wooldridge sandwich), and point estimates match the `arch` Python package to three decimals. Code: `volatility_ps.py`.

	Mean	S.d.	Ann. vol	Skew	Excess kurt.	Min	Max
Market	0.044	1.13	17.9%	−0.32	11.0	−11.8	11.5
MSFT	0.102	2.02	32.1%	0.23	6.5	−15.6	19.6

Question 1

(a) Autocorrelations of returns and squared returns

Prior. The market should show stronger volatility clustering. Persistent volatility is economy-wide (recessions, crises, policy uncertainty), and the index carries all of it. MSFT adds firm-specific news (earnings, antitrust rulings) that arrives as one-day jumps that do not persist. Those jumps add non-persistent noise to r_t^2 and lower its autocorrelation. Returns themselves should be nearly unpredictable for both series.

Assessment: the prior is correct. Squared (demeaned) returns are autocorrelated at all 20 lags for both series (Ljung–Box $Q(20)$ of 11,056 for the market and 1,853 for MSFT). The market's autocorrelations are higher, averaging 0.256 against 0.103 over lags 1–20. They also decay more slowly: at lag 20 the market keeps 57% of its lag-1 value (0.176 vs 0.311), while MSFT keeps 26% (0.048 vs 0.182). Absolute returns, which are less sensitive to outliers, give the same ranking. The jump mechanism is visible in the data: MSFT's largest moves (+19.6% on 19 Oct 2000, −14.5% on 3 Apr 2000, −15.6% on 24 Apr 2000) coincided with market moves of less than 4%. Return autocorrelations are all below 0.06 in absolute value. Against the heteroskedasticity-robust band, only 4 of 20 market lags and 1 of 20 MSFT lags are significant.

(b) Gaussian GARCH(1,1) by maximum likelihood

Model.

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \quad z_t \overset{iid}{\sim} N(0,1), \qquad \sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2,$$

with $\omega > 0$, $\alpha, \beta \geq 0$ and $\alpha + \beta < 1$.

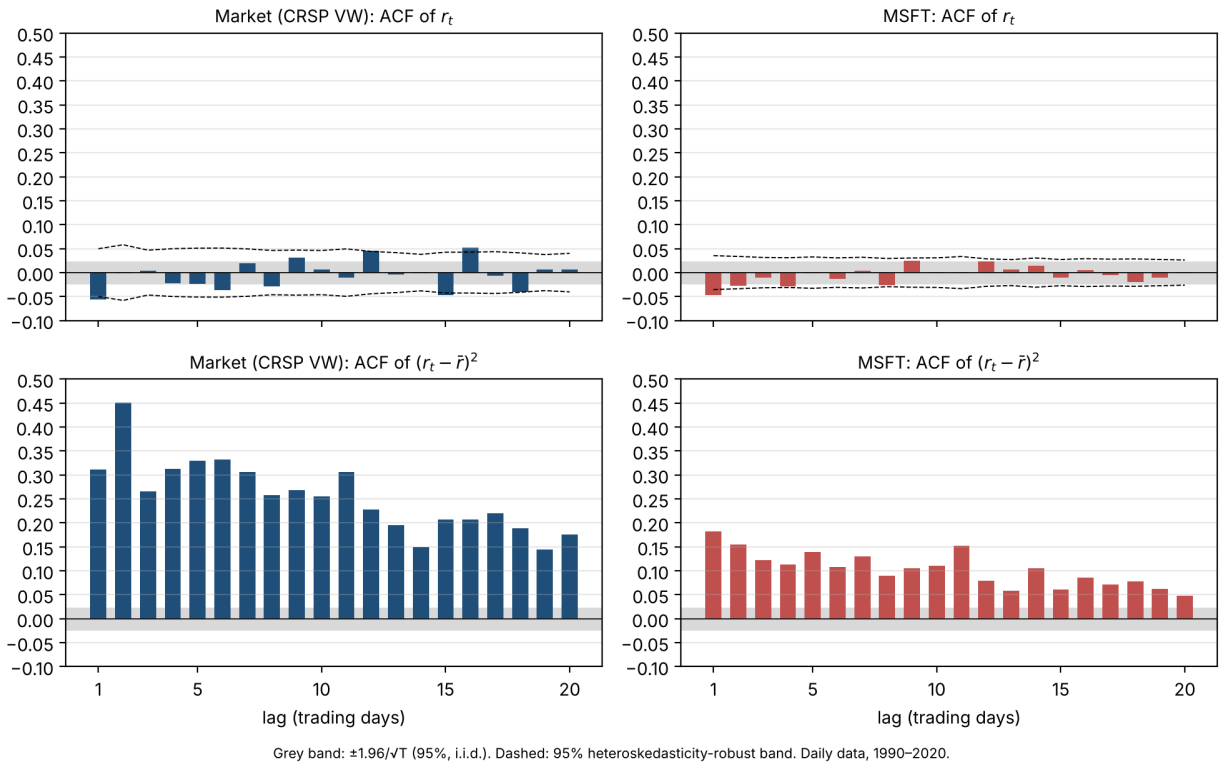


Figure 1: Autocorrelations at lags 1–20. Grey band: i.i.d. 95% band $\pm 1.96/\sqrt{T}$. Dashed: heteroskedasticity-robust 95% band.

Log likelihood. Since $r_t \mid \mathcal{F}_{t-1} \sim N(\mu, \sigma_t^2)$, with $\theta = (\mu, \omega, \alpha, \beta)$,

$$\mathcal{L}(\theta) = -\frac{1}{2}\sum_{t=1}^{T}\left[\log(2\pi) + \log\sigma_t^2(\theta) + \frac{(r_t-\mu)^2}{\sigma_t^2(\theta)}\right], \qquad \hat{\theta} = \arg\max_{\theta}\mathcal{L}(\theta),$$

where $\sigma_t^2(\theta)$ follows the recursion from $\sigma_1^2 = T^{-1}\sum_t (r_t - \mu)^2$.

	μ	ω	α	β	$\alpha+\beta$	Half-life	Long-run vol	$\log L$
Market	0.071	0.0175	0.112	0.873	0.985	47	17.4%	−10,110.2
	(0.009)	(0.0035)	(0.012)	(0.012)	(0.005)			
MSFT	0.116	0.066	0.079	0.906	0.985	47	33.9%	−15,649.6
	(0.018)	(0.036)	(0.025)	(0.030)	(0.009)			

Table 1: GARCH(1,1) estimates; robust s.e. in parentheses. The s.e. of $\alpha+\beta$ uses the delta method with the full covariance matrix. Half-life (days) $= \log 0.5/\log(\alpha+\beta)$ for deviations of the variance from its long-run level. Long-run vol $= \sqrt{252\,\omega/(1-\alpha-\beta)}$.

Both series have small α, large β and $\alpha + \beta = 0.985$, so a deviation of the variance from its long-run level halves in about 47 trading days. Because $\hat{\alpha}$ and $\hat{\beta}$ are strongly negatively correlated (-0.93 market, -0.97 MSFT), the s.e. of the sum must include their covariance: it is 0.0046 for the market and 0.0086 for MSFT. A one-sided test of $\alpha + \beta = 1$ rejects for the market ($t = 3.2$) but only marginally for MSFT ($t = 1.7$, $p \approx 0.045$). The market reacts more to news ($\alpha = 0.112$ vs 0.079), which matches its stronger squared-return autocorrelation. The implied long-run volatilities

(17.4%, 33.9%) match the sample values (17.9%, 32.1%). The model removes the clustering: $Q(20)$ of the squared standardized residuals is 25.2 ($p = 0.19$) for the market and 11.6 ($p = 0.93$) for MSFT. It does not remove the fat tails (residual kurtosis 4.9 and 8.8), so the Gaussian likelihood is a quasi-likelihood. The estimates remain consistent, but robust standard errors are required; for MSFT they are 2–3 times the classical ones.

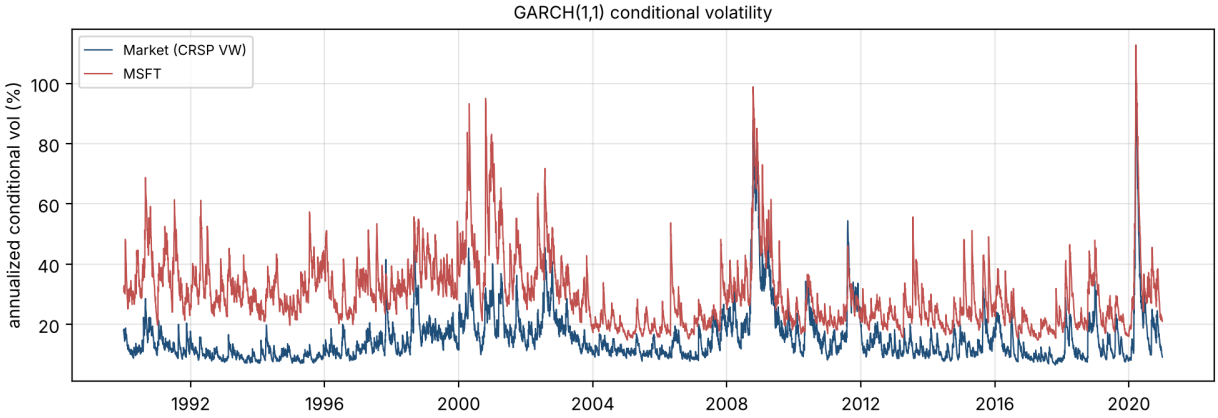


Figure 2: GARCH(1,1) conditional volatility, annualized.

(c) Is MSFT's volatility just the market moving?

No. In the market model $r_{MSFT,t} = a + b\, r_{m,t} + u_t$, $\hat{b} = 1.15$ (White s.e. 0.021) and $R^2 = 0.41$. The market explains 41% of MSFT's daily variance; firm-specific news explains 59%.

The firm-specific part also has its own dynamics. GARCH(1,1) on u_t gives $\alpha = 0.024$ (0.006) and $\beta = 0.971$ (0.007), with persistence 0.995 (0.002), and $Q(20)$ of u_t^2 is 704. Firm-specific volatility therefore clusters on its own and is at least as persistent as market volatility.

Combining the two GARCH fits gives an *approximate* market-related variance share

$$s_t = \frac{b^2 \hat{\sigma}^2_{m,t}}{b^2 \hat{\sigma}^2_{m,t} + \hat{\sigma}^2_{u,t}}.$$

It is approximate because it assumes a constant beta and a market return that is conditionally uncorrelated with u_t; OLS imposes only unconditional orthogonality, and yearly betas range from 0.73 to 2.30. The implied total variance averages 1.05 times MSFT's own GARCH variance (10th–90th percentile 0.72–1.43). The share averages 0.36 and exceeds one half on only 20% of days.

The claim holds only in market-wide crises (shares of 0.64 in 2008–09 and 0.87 in March–April 2020), when market volatility spikes and firm-specific volatility does not. It fails in calm periods (0.32 in 2017) and in MSFT-specific episodes such as 2000–02, when firm-specific volatility reached 33.8%. The yearly R^2 has risen to 0.64–0.75 in 2018–20 as Microsoft became a large index weight. Over 1990–2020 as a whole, however, most of MSFT's variance is firm-specific.

Period	Approx. market share	Market-related vol	Firm-specific vol
Full sample 1990–2020	0.36	17.8%	23.5%
Dot-com and antitrust 2000–02	0.36	25.3%	33.8%
Financial crisis Sep 2008–Jun 2009	0.64	49.4%	33.7%
COVID crash Mar–Apr 2020	0.87	71.3%	24.2%
Calm year 2017	0.32	10.0%	14.7%

Table 2: Average approximate market-related share of MSFT's conditional variance, and the average annualized volatility of each part.

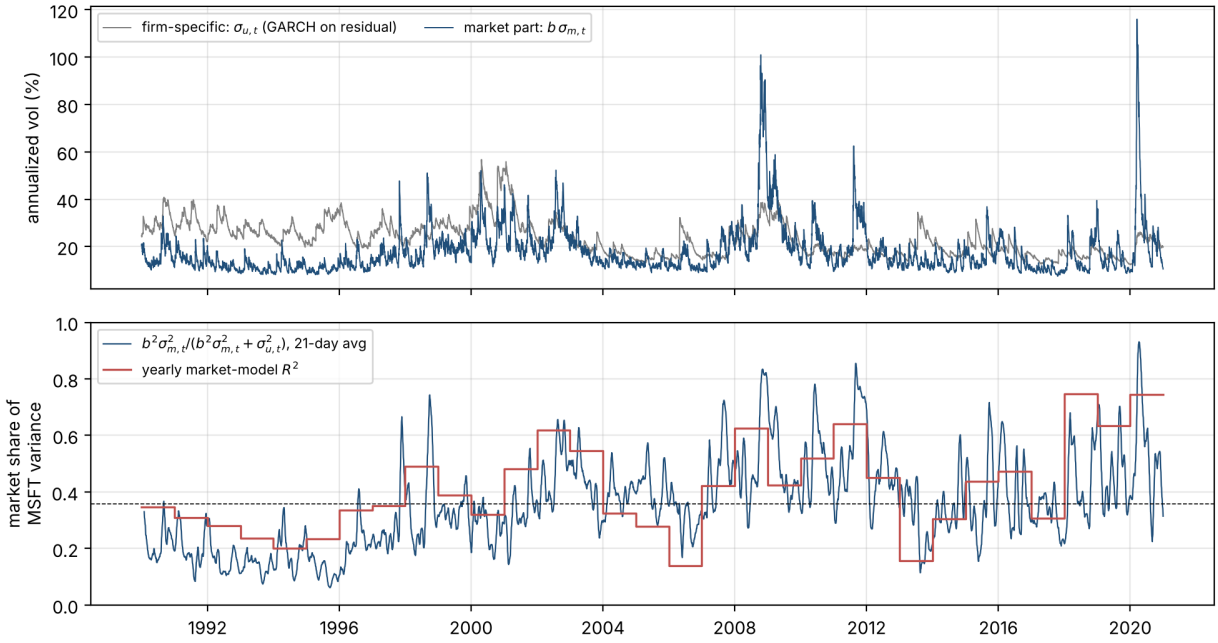


Figure 3: Top: market-related ($b\,\sigma_{m,t}$) and firm-specific ($\sigma_{u,t}$) volatility of MSFT. Bottom: approximate market-related share of MSFT's variance (dashed: mean) and yearly market-model R^2.

(d) Market risk over the next month (as of 31 Dec 2020)

The h-day variance forecast is $\mathbb{E}_t\sigma^2_{t+h} = \bar{\sigma}^2 + (\alpha+\beta)^{h-1}(\sigma^2_{t+1} - \bar{\sigma}^2)$, and the one-month variance is the sum of 21 such forecasts (one-month s.d. 3.0%, against 5.0% at the long-run level). The ranges and loss probabilities come from 200,000 simulated 21-day paths of the fitted model with Gaussian daily shocks, because the monthly GARCH return is not exactly Gaussian.

- **Risk is low right now.** The model estimates market volatility at about 9% a year, half its long-run level of 17%. The end of 2020 was quiet, and quiet periods tend to persist.
- **It will drift back up slowly.** The gap to normal, measured in variance, halves about every two months, so the forecast rises only to about 12% by late January. A typical monthly move is ±3% (against ±5% normally). In 19 of 20 simulated months the market ends between −4.5% and +7.5%, and a loss worse than 6% occurs about 1 month in 100.
- **This can change overnight.** A single large shock raises the forecast immediately: on

17 March 2020 the same model put volatility near 100% a year. The model also assumes bell-curve shocks, so it understates crash risk.

Question 2 (PhD): Gaussian GARCH-in-mean

MLE objective. With $r_t = \mu + \lambda\, g(\sigma_t^2) + \varepsilon_t$, $\varepsilon_t = \sigma_t z_t$, $z_t \sim N(0,1)$, $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, and $g(\sigma^2) = \sigma^2$ (or σ, $\log\sigma^2$),

$$\hat{\theta} = \arg\max_{\theta} \; -\frac{1}{2}\sum_{t=1}^{T}\left[\log 2\pi + \log\sigma_t^2(\theta) + \frac{\left(r_t - \mu - \lambda g(\sigma_t^2(\theta))\right)^2}{\sigma_t^2(\theta)}\right] \quad \text{s.t. } \omega > 0,\ \alpha,\beta \geq 0,\ \alpha+\beta < 1.$$

Because ε_t now depends on the variance parameters, the two recursions must be run jointly (σ_t^2 from ε_{t-1}, then ε_t from σ_t^2). The information matrix is no longer block-diagonal, so the mean and variance parameters must be estimated jointly. (For illustration, with $g = \sigma^2$: $\hat{\lambda} = 0.041$ (0.013) for the market and 0.012 (0.009) for MSFT.)

Return predictors X_t. Replace the mean by $m_t = \mu + \lambda g(\sigma_t^2) + \gamma' X_{t-1}$ and use $\varepsilon_t = r_t - m_t$ in the same objective; the variance recursion is then fed this residual. X_{t-1} must be known at $t-1$ and must not be collinear with $g(\sigma_t^2)$.

Variance predictors Q_t. Use $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 + \delta' Q_{t-1}$ in the same objective, with four additional conditions. (i) *Positivity*: $\sigma_t^2 > 0$ is no longer implied by $\omega, \alpha, \beta \geq 0$. Impose $Q_t \geq 0$ and $\delta \geq 0$, restrict $\omega + \delta' Q_t > 0$ on the support of Q, or reparametrize (e.g. $\exp(\delta' Q_{t-1})$ or an EGARCH-X model for $\log\sigma_t^2$). (ii) *Predetermined*: $Q_{t-1} \in \mathcal{F}_{t-1}$, or σ_t^2 is not a conditional variance. (iii) *Stationarity*: besides $\alpha + \beta < 1$, Q_t must be strictly stationary and ergodic with $\mathbb{E}|Q_t| < \infty$, so that $\mathbb{E}\sigma^2 = (\omega + \delta'\mathbb{E}Q)/(1-\alpha-\beta)$ exists; a trending Q makes σ_t^2 non-stationary. With identification, these conditions give consistency. Asymptotic normality additionally needs an interior true parameter, $\mathbb{E}z_t^4 < \infty$, and sufficient moments of Q_t and σ_t^2 (Francq and Zakoïan, 2004; Han and Kristensen, 2014). (iv) *Identification*: Q must vary over time and must not be collinear with ε_{t-1}^2 or σ_{t-1}^2. For testing: if $\delta \geq 0$ is imposed, $\delta = 0$ lies on the boundary, and for a scalar δ with the other parameters interior the LR statistic converges to $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ rather than χ_1^2 (Andrews, 2001).

Question 3 (PhD): GJR-GARCH(1,1,1) with Student-t innovations

Model. $r_t = \mu + \varepsilon_t$, $\varepsilon_t = \sigma_t z_t$, with z_t i.i.d. standardized t_ν (mean 0, variance 1, $\nu > 2$), and

$$\sigma_t^2 = \omega + (\alpha + \gamma\,\mathbf{1}\{\varepsilon_{t-1} < 0\})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2.$$

The log likelihood is

$$\mathcal{L} = \sum_t \left[\log\Gamma\left(\tfrac{\nu+1}{2}\right) - \log\Gamma\left(\tfrac{\nu}{2}\right) - \tfrac{1}{2}\log\left(\pi(\nu-2)\right) - \tfrac{1}{2}\log\sigma_t^2 - \tfrac{\nu+1}{2}\log\left(1 + \frac{\varepsilon_t^2}{(\nu-2)\sigma_t^2}\right)\right].$$

Non-integration condition. Since $\varepsilon_{t-1}^2 = \sigma_{t-1}^2 z_{t-1}^2$ and $\mathbf{1}\{\varepsilon_{t-1} < 0\} = \mathbf{1}\{z_{t-1} < 0\}$,

$$\sigma_t^2 = \omega + \sigma_{t-1}^2\left[\alpha z_{t-1}^2 + \gamma z_{t-1}^2 \mathbf{1}\{z_{t-1} < 0\} + \beta\right].$$

z_{t-1} is independent of $\sigma^2_{t-1} \in \mathcal{F}_{t-2}$, $\mathbb{E}z^2 = 1$, and by symmetry of the standardized t, $\mathbb{E}[z^2\mathbf{1}\{z < 0\}] = \frac{1}{2}$. Hence

$$\mathbb{E}_{t-2}\sigma^2_t = \omega + \phi\,\sigma^2_{t-1}, \qquad \phi = \alpha + \tfrac{1}{2}\gamma + \beta,$$

and iterating gives $\mathbb{E}_{t-k}\sigma^2_t = \omega\sum_{j=0}^{k-2}\phi^j + \phi^{k-1}\sigma^2_{t-k+1}$. Variance shocks die out and $\mathbb{E}\varepsilon^2_t = \omega/(1-\phi)$ is finite if and only if

$$\boxed{\alpha + \tfrac{1}{2}\gamma + \beta < 1} \qquad (\omega > 0,\ \alpha \geq 0,\ \alpha + \gamma \geq 0,\ \beta \geq 0,\ \nu > 2).$$

At $\phi = 1$ the variance is integrated: shocks are permanent and the unconditional variance does not exist.

	μ	ω	α	γ	β	ν	ϕ	$\log L$
Market	0.065	0.0137	0.009	0.163	0.895	6.9	0.985	−9,832.4
	(0.008)	(0.0024)	(0.006)	(0.018)	(0.010)	(0.57)	(0.003)	
MSFT†	0.073	0.032	0.056	0.068	0.910	5.3	0.99999	−15,220.2
	(0.016)	(0.010)	(0.010)	(0.015)	(0.014)	(0.32)	–	

Table 3: GJR-GARCH-t estimates; robust s.e. in parentheses. †The constraint $\phi \leq 1 - 10^{-5}$ imposed in estimation binds for MSFT ($\hat{\phi} = 0.99999$). Without it $\hat{\phi} = 1.0017$ with a log-likelihood gain of 0.12 (LR = 0.24). The MSFT standard errors are computed as if the estimate were interior and are only indicative.

	GARCH-N	GARCH-t	GJR-N	GJR-t
Market	−10,110.2	−9,927.5	−9,999.7	−9,832.4
MSFT	−15,649.6	−15,238.1	−15,624.0	−15,220.2

Table 4: Log likelihood by model: which feature (fat tails or leverage) improves the fit.

Comparison with GARCH(1,1). GJR-t raises the log likelihood by 278 for the market and 429 for MSFT. Fat tails account for most of the gain ($\hat{\nu} = 6.9$ and 5.3; +183 and +412 from t innovations alone). Leverage adds +111 for the market but only +25 for MSFT. For the market, the symmetric α collapses to 0.009 (insignificant) and $\gamma = 0.163$ carries the ARCH effect. Its persistence is unchanged ($\phi = 0.985$, s.e. 0.0035, $t = 4.3$ against one), and its long-run volatility falls from 17.4% to 15.2%. For MSFT, $\hat{\phi}$ sits on the constraint, so its variance is indistinguishable from an integrated process and its long-run variance is not identified.

Leverage: market vs MSFT. Leverage is much stronger for the market. Bad news raises the market's next-day variance 19 times as much as good news ($\alpha+\gamma = 0.171$ vs $\alpha = 0.009$), against 2.2 times for MSFT (0.124 vs 0.056). MSFT's γ is 0.42 times the market's, close to the market-related share of MSFT's variance (0.36–0.41). This argues against financial leverage, since Microsoft had essentially no debt, and for a systematic channel: volatility feedback and correlation spikes in selloffs. MSFT's firm-specific news raises its volatility whether the news is good or bad.

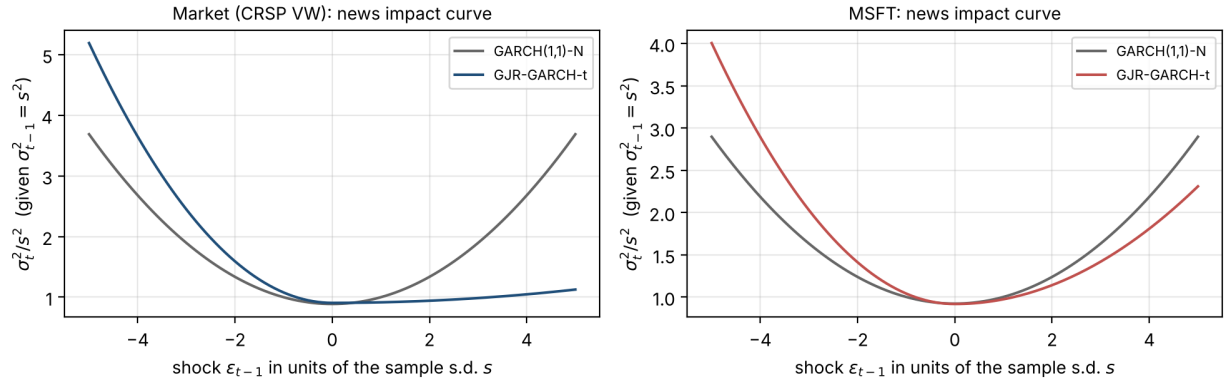


Figure 4: News impact curves: next-day variance after a shock ε_{t-1}, given that the previous variance equals the sample variance s^2. Axes are scaled by the sample s.d. because the long-run variance does not exist for MSFT's GJR-t estimate.